

生态研究前沿速览_2026-06-06

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2026-06-06

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保护与生物多样性

1. Widespread range contraction of carnivores in protected areas of China
(中国保护区内食肉动物分布范围广泛收缩)

Nature Sustainability | 2026-05-27 | 【Recent】



图：Graphical abstract / article figure source: <https://www.nature.com/articles/s41893-026-01855-2/figures/1>

Abstract

The study integrates historical and contemporary wildlife survey records to test whether China's protected areas have prevented range losses among medium and large mammals. It reports widespread contraction of carnivore distributions inside protected areas, showing that formal protection alone has not fully offset habitat loss, disturbance and human-wildlife conflict.

中文摘要

研究整合历史调查和 2008-2021 年现状资料，系统评估中国自然保护区对大中型哺乳动物，尤其是食肉动物的保护成效。结果显示，许多食肉动物即使在保护区内仍发生分布范围收缩，说明保护地边界并不等同于有效保护。

深度解读

这篇文章直接触及中国保护地体系的核心问题：面积扩张之后，物种是否真正留在保护区内。大型食肉动物对栖息地连通性、人类干扰和猎物基础都高度敏感，因此是保护质量的早期预警指标。研究提示下一步应从“划区”转向“提升管理有效性”，包括生态廊道、社区冲突治理和长期监测。对国家公园和 30x30 目标评估也具有直接政策意义。

DOI: <https://doi.org/10.1038/s41893-026-01855-2>

环境与气候

2. Atmospheric evaporative demand has increased drought severity worldwide

(大气蒸发需求上升加剧全球干旱严重程度)

Nature | 2026-06-04 | **【Latest】**



图：Graphical abstract / article figure source: <https://www.nature.com/articles/s41586-026-08999-5/figures/1>

Abstract

This analysis attributes a large share of recent drought intensification to rising atmospheric evaporative demand. By separating precipitation deficits from the atmosphere's drying power, it clarifies why drought impacts can increase even where rainfall trends are weak or spatially mixed.

中文摘要

研究区分降水亏缺与大气“吸水能力”对干旱的贡献，指出升温驱动的大气蒸发需求上升已经显著放大干旱强度。即使部分地区降水趋势不明显，植被、土壤和河流仍可能因蒸发压力增加而承受更强水分胁迫。

深度解读

传统干旱监测容易过度依赖降水异常，而该文强调温度、湿度、风速和辐射共同决定生态系统失水压力。它对森林死亡、草地退化、农业灌溉和流域生态流量都有直接意义。未来干旱预警需要把大气蒸发需求纳入核心指标，否则会低估暖干复合事件的生态风险。

DOI: <https://doi.org/10.1038/s41586-026-08999-5>

保护与生物多样性

3. Improved forest protection at Chinese heritage relic sites

(中国遗产遗迹地提升森林保护成效)

Nature Sustainability | 2026-05-22 | 【Recent】



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Rapid deforestation and accelerating human pressures throughout the twentieth century reshaped forest landscapes worldwide. Using a national database of 363,282 heritage relic sites in China, historical forest coverage data and a high-resolution human footprint index, the authors show that heritage relics can function as long-term forest refugia. Heritage trees and traditional villages are associated with higher forest coverage and faster regeneration, especially in southern China and lower-pressure regions, but the effect weakens as human footprints intensify.

中文摘要

研究整合中国 363,282 个遗产遗迹点、1900-2020 年历史森林覆盖数据和 2020 年高分辨率人类足迹指数，发现遗产树和传统村落等生物文化遗产可作为长期森林避难所，在毁林时期维持森林、在大规模恢复时期促进再生。其保护效应在华南和人类压力较低区域更明显，但会随人类足迹增强而减弱。

深度解读

这项研究把文化遗产保护与森林生态韧性直接连接起来，说明传统村落、古树和宗教文化空间并非只是文化资产，也可能是长期维持生态系统服务的基础设施。它为中国及全球生物多样性保护提供了低成本、可嵌入国土空间治理的路径。关键限制在于保护效应依赖周边人类压力，因此遗产保护若不与土地利用管控联动，生态收益会被快速削弱。

DOI: <https://doi.org/10.1038/s41893-026-01846-3>

进化与动物

4. Signatures of gene transfer in the parallel evolution of osmotrophic specialization in eukaryotes

(基因转移塑造真核生物渗透营养专化的平行演化)

Nature Ecology & Evolution | 2026-05-25 | 【Recent】



图：Graphical abstract / article figure source: <https://www.nature.com/articles/s41559-026-03054-w/figures/1>

Abstract

Comparative genomics across 86 eukaryotic species reconstructs four independent transitions towards osmotrophy. The authors identify a shared metabolic toolkit and infer that horizontal gene transfer contributed substantially to this convergent trophic specialization.

中文摘要

研究比较 86 个真核物种基因组，重建四个类群独立转向渗透营养的演化历史。结果显示，这些类群拥有共同富集的代谢功能工具箱，其中相当一部分信号与水平基因转移相符。

深度解读

该文把生态位转变、营养方式演化和基因组机制连接起来，解释了远缘真核生物为何能反复演化出相似的吸收型营养策略。对微生物生态而言，水平基因转移不仅改变单个性状，也可能重塑整个营养功能组。它还提醒我们，在理解分解者和寄生/腐生类群演化时，物种树并不能完全代表功能演化路径。

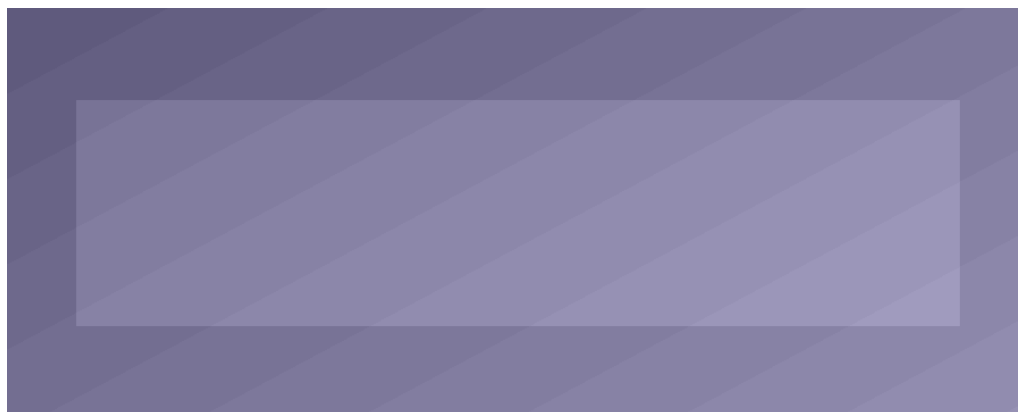
DOI: <https://doi.org/10.1038/s41559-026-03054-w>

环境与气候

5. Increased sensitivity of the Antarctic Ice Sheet to decreasing CO2 across the Mid-Pleistocene Transition

(中更新世转型后南极冰盖对 CO₂ 下降的敏感性增强)

Nature Geoscience | 2026-05-28 | 【Recent】



图：Graphical abstract / article figure source: <https://www.nature.com/articles/s41561-026-01979-2/figures/1>

Abstract

Ice-sheet simulations driven by realistic climate fields over the past three million years identify a nonlinear Antarctic regime shift around the Mid-Pleistocene Transition. The results indicate much higher ice-sheet sensitivity below a CO₂ threshold near 240 ppmv, with implications for interpreting threshold behaviour in future sea-level projections.

中文摘要

研究将三百万年古气候场输入冰盖-冰架模型，发现南极冰盖在中更新世转型前后出现非线性机制变化。当大气 CO₂ 低于约 240 ppmv 时，冰盖对气候和海平面强迫的响应显著增强。

深度解读

该文的重要性在于用长时间尺度模拟揭示冰盖系统的阈值行为。虽然研究对象是过去的低 CO₂ 冰期状态，但它强调冰盖并非线性响应气候强迫。对未来海平面风险评估而言，识别冰-海洋-基岩反馈的阈值，比只提高模型分辨率更关键。

DOI: <https://doi.org/10.1038/s41561-026-01979-2>

保护与生物多样性

6. Human shielding from wolves facilitates jackal expansion across Europe

(人类庇护效应帮助金豺避开狼并扩张至欧洲)

Nature Ecology & Evolution | 2026-05-25 | 【Recent】



图：Graphical abstract / article figure source: <https://www.nature.com/articles/s41559-026-03060-y/figures/1>

Abstract

Playback surveys and distribution modelling show that grey wolves are a major constraint on golden jackal occurrence, but human presence weakens this suppression by providing refuges near people. The model projects continued jackal expansion across Europe under combined human shielding, warming and landscape change.

中文摘要

研究利用欧洲尺度回放调查和分布模型发现，灰狼是限制金豺出现的关键因素，但人类活动区会削弱狼对金豺的压制，使金豺获得靠近人类的避难空间。模型显示，气候变暖、景观改变和人类庇护效应将共同推动金豺继续扩张。

深度解读

这项研究展示了人类活动如何间接改变捕食者之间的相互作用。顶级捕食者恢复通常被视为生态修复利好，但如果人类空间提供中型捕食者避难所，级联效应会变得更复杂。它对欧洲野生动物管理的启示是，不能只看单个物种恢复，还要同时评估人类活动对物种间关系的重构。

DOI: <https://doi.org/10.1038/s41559-026-03060-y>

植物与农业

7. Global rice paddy greenhouse gas emissions have doubled over the past six decades driven by area expansion and intensified residue incorporation

(面积扩张和秸秆还田强化推动全球稻田温室气体排放六十年翻倍)

Nature Food | 2026-05-22 | 【Recent】



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Using a data-driven model, a process-based ecosystem model and a meta-analysis of more than 1,255 field experiment sites, this study estimates that net greenhouse gas emissions from global rice paddies roughly doubled from 1961-1980 to 2001-2020. The increase was mainly driven by higher soil CO₂ and CH₄ emissions, linked to rice-area expansion and intensified residue incorporation.

中文摘要

研究使用数据驱动模型、过程生态系统模型和 1,255 多个田间试验点的荟萃分析，估算全球稻田净温室气体排放从 1961-1980 年到 2001-2020 年约翻倍。增长主要来自土壤 CO₂ 和 CH₄ 排放上升，与稻田面积扩张和秸秆还田强化密切相关。

深度解读

稻田既是粮食安全核心系统，也是甲烷和二氧化碳排放的重要来源。该研究的价值在于用多方法交叉约束长期排放趋势，减少单一模型偏差。未来水稻低碳管理需要在粮食产量、秸秆资源化、间歇灌溉和土壤碳管理之间寻找区域化组合，而不是简单压缩稻作。

DOI: <https://doi.org/10.1038/s43016-026-01355-8>

海洋与水生

8. Critical dependence of global ocean heat monitoring on the ocean observing system

(全球海洋热监测高度依赖海洋观测系统)

Nature Climate Change | 2026-05-22 | 【Recent】



图：Graphical abstract / article figure source: <https://www.nature.com/articles/s41558-026-02661-6/figures/1>

Abstract

The study quantifies how losses in the global ocean observing system degrade estimates of ocean heat content. It shows that the system is highly vulnerable to geographically concentrated data gaps, making sustained international observations essential for climate monitoring, forecasts and ocean risk management.

中文摘要

研究评估全球海洋观测系统数据缺失对海洋热含量估计的影响，发现观测损失会迅速降低年度海洋增暖评估精度，且特定国家或区域观测缺口造成的损害可能超过随机缺失。维持全球协调观测对于气候监测、预报和海洋管理至关重要。

深度解读

海洋热含量是地球能量失衡和海平面变化的核心指标，但它依赖长期、跨国、空间覆盖均衡的观测网络。该文把观测系统本身作为气候基础设施来量化风险，具有很强的政策意义。对生态研究而言，海温、海洋热浪、渔业资源和珊瑚礁风险都依赖这些基础观测。

DOI: <https://doi.org/10.1038/s41558-026-02661-6>

环境与气候

9. A multidimensional assessment of systemic cooling poverty in the global south

(全球南方系统性降温贫困的多维评估)

Nature Sustainability | 2026-05-22 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Extreme heat increasingly threatens human health where climate exposure intersects with social, infrastructural and institutional deprivation. Combining household survey data from 1,155,106 households, climate records and infrastructure datasets, the study builds a systemic cooling poverty index across 28 mostly global-south countries. The analysis finds that almost 600 million people experience high deprivation across cooling-related dimensions.

中文摘要

研究结合 28 个以全球南方为主国家的 1,155,106 户家庭调查、气候记录和基础设施数据，构建系统性降温贫困指数，覆盖气候暴露、基础设施与资产、社会 and 热不平等、健康、教育与工作标准五个维度。结果显示，近 6 亿人处于较高降温贫困水平。

深度解读

该文将高温适应从单一空调可及性扩展为社会-基础设施-健康复合问题。它提示热风险治理不能只依赖能源消费扩张，否则可能加剧排放和不平等。对城市气候适应而言，遮阴、住房质量、公共冷却空间、劳动保护和电力系统韧性需要被纳入同一政策框架。

DOI: <https://doi.org/10.1038/s41893-026-01845-4>

城市生态与方法

10. Synthetic control methods enable stronger causal inference using participatory science data in cities

(合成控制方法增强城市参与式科学数据的因果推断能力)

Nature Ecology & Evolution | 2026-05-20 | 【Recent】



图：Graphical abstract / article figure source: <https://www.nature.com/articles/s41559-026-03084-4/figures/1>

Abstract

The paper combines iNaturalist records with synthetic-control methods to infer urban biodiversity responses to disturbance. In a Philadelphia case study, the approach detected a post-Hurricane Ida decline in research-grade bee observations that several conventional ecological designs failed to detect.

中文摘要

研究将 iNaturalist 公众科学记录与合成控制方法结合，用于估计城市生物多样性对气候事件和政策干预的响应。在费城飓风 Ida 案例中，该方法识别出灾后两年蜜蜂观测显著下降，而传统分析方法未能检出。

深度解读

公众科学数据规模大但偏差复杂，过去常被限制在描述性监测。该文的贡献在于为城市生态学提供可操作的准实验框架，使非随机观测数据更接近因果评估。未来评估极端天气、绿地政策和城市更新对物种的影响时，这类方法会显著提高证据质量。

DOI: <https://doi.org/10.1038/s41559-026-03084-4>

环境与气候

11. Glacial forefields shift from carbon sinks to greenhouse-gas sources after deglaciation

(冰川退缩后前缘地带可能由碳汇转为温室气体源)

Communications Earth & Environment | 2026-06-05 | 【Latest】



图：Graphical abstract / article figure source: <https://www.nature.com/articles/s43247-026-03600-0/figures/1>

Abstract

The paper examines how newly exposed glacier forefields change carbon processing as meltwater influence declines and soils develop. It suggests that some deglaciated landscapes can move from net CO₂ uptake toward methane and CO₂ production, creating a warming feedback.

中文摘要

研究关注冰川消退后新暴露地表的碳循环变化，指出随着融水影响减弱和土壤发育，部分冰川前缘地带可能从净吸收 CO₂ 转向释放 CH₄ 和 CO₂。这意味着冰川退缩不仅改变水文，也会改变陆地-大气碳交换。

深度解读

冰川退缩通常被讨论为水资源和海平面问题，但其暴露地表也在快速形成新的微生物和植被生态系统。该文强调早期演替阶段的生物地球化学反馈，尤其是湿润、缺氧微环境中的甲烷风险。高山和极地生态监测需要把温室气体通量纳入退冰区长期样地。

环境与气候

12. Incorporating air quality health impacts into the social cost of carbon

(将空气质量健康影响纳入碳社会成本)

Nature Climate Change | 2026-05-28 | **【Recent】**



图：Graphical abstract / article figure source: <https://www.nature.com/nclimate/articles?year=2026>

Abstract

This review argues that climate-driven air-pollution damages remain insufficiently represented in social cost of carbon models. It lays out evidence and modelling needs for integrating air-quality health impacts into climate-damage valuation.

中文摘要

综述指出，气候变化会通过臭氧、颗粒物和气象条件变化影响空气污染健康负担，但这些损害在现有碳社会成本模型中仍纳入不足。文章梳理证据基础，并提出将空气质量健康影响整合进气候损害估值的建模框架。

深度解读

碳社会成本是气候政策定价的关键参数，漏计空气质量健康损害会低估减排收益。该文的重要性不在单个生态过程，而在环境健康与气候经济评估的接口。对生态环境管理来说，它支持把空气污染协同效益纳入碳政策、能源转型和城市规划评价。

DOI: <https://doi.org/10.1038/s41558-026-02669-y>

海洋与水生

13. Ecological and genomic signatures of the convergent evolution of planktivory in fossil and living reef fishes over deep time

(深时尺度上化石与现生礁鱼浮游生物食性的趋同进化生态与基因组信号)

Nature Communications | 2026-05-22 | **【Recent】**



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

This study examines ecological and genomic signatures of convergent evolution of planktivory in fossil and living reef fishes over deep time. By linking fossil evidence with living reef-fish genomic and ecological patterns, it provides a long-term view of how reef fish trophic strategies repeatedly emerged.

中文摘要

研究结合化石和现生礁鱼资料，追踪浮游生物食性在深时尺度上的趋同演化，并从生态和基因组层面识别相关信号。该工作为理解珊瑚礁鱼类营养策略如何反复出现提供了长期演化视角。

深度解读

珊瑚礁鱼类功能多样性是维持礁生态系统稳定的重要基础。把化石记录与现生基因组结合，可以区分短期生态适应和长期演化约束。该研究也提示，在气候变化和珊瑚礁退化背景下，评估功能群脆弱性需要考虑其演化可塑性和历史路径。

DOI: <https://doi.org/10.1038/s41467-026-73110-3>

城市生态与环境

14. Spatiotemporal changes in inter-city sustainability impacts linked to emission challenges worldwide

(全球排放挑战关联的城市间可持续影响时空变化)

Nature Communications | 2026-05-29 | 【Recent】



图：Graphical abstract / article figure source: <https://www.nature.com/articles/s41467-026-73515-0/figures/1>

Abstract

The paper analyses how emission-related sustainability impacts propagate across interconnected city systems over time. It highlights that urban emissions and air-pollution burdens are not only local outcomes but are shaped by inter-city links, supply chains and coordination failures.

中文摘要

研究从全球城市网络视角分析排放相关可持续影响的时空变化，指出温室气体和空气污染负担不仅发生在本地，也会通过城市间联系和供应链外溢。城市协同治理可降低孤立行动带来的效率损失。

深度解读

城市可持续性评估常把城市当成独立单元，但现实中的排放、消费和污染影响高度网络化。该研究强调跨城市协同，而不是只优化单城指标。对区域生态环境治理而言，这意味着碳减排、空气质量和产业转移政策需要放在城市群和全球供应链尺度上评价。

DOI: <https://doi.org/10.1038/s41467-026-73515-0>

植物与农业

15. Metabolic feedbacks drive population dynamics and can lead to oscillations among leaf bacteria

(代谢反馈驱动叶际细菌种群动态并可产生振荡)

Nature Communications | 2026-05-29 | 【Recent】



图：Graphical abstract / article figure source: <https://www.nature.com/articles/s41467-026-73548-5/figures/1>

Abstract

Using microfluidics and computational modelling, the study links plant-associated bacterial interactions to molecular metabolic feedbacks. It shows how feedbacks among leaf bacteria can generate emergent population dynamics, including oscillations.

中文摘要

研究结合微流控实验和计算模型，解析植物叶际细菌之间的代谢互作如何影响种群动态。结果表明，微生物代谢反馈可产生复杂的群落动态，甚至导致种群振荡。

深度解读

叶际微生物群落影响植物健康、病害抗性和养分循环，但其动态机制长期难以量化。该文把微尺度代谢互作与群落水平动态连接起来，有助于解释为什么同一植物表面的微生物组成会快速波动。它也为设计更稳定的植物有益菌群提供机制基础。

DOI: <https://doi.org/10.1038/s41467-026-73548-5>

海洋与水生

16. Multitasking sponges

(多任务海绵：礁区海绵的混合营养与碳预算意义)

Nature Ecology & Evolution | 2026-05-28 | 【Recent】



图：Graphical abstract / article figure source: <https://doi.org/10.1111/1365-2435.70311>

Abstract

This research highlight reports new evidence that many Caribbean reef sponges are mixotrophic and can meet part or all of their respiratory demand through photosymbiont-supported production. Scaled to reefs, sponge gross primary production may be large enough to alter benthic carbon budgets.

中文摘要

该研究亮点评述 Functional Ecology 论文，指出加勒比礁区许多海绵并非单纯异养，而是借助蓝细菌或甲藻共生体进行光合生产，部分物种可覆盖相当比例的呼吸需求。尺度外推显示，海绵初级生产可能被现有珊瑚礁碳预算低估。

深度解读

在珊瑚退化和海绵相对占优的背景下，海绵功能角色需要重新评价。它们不仅过滤水体，也可能作为重要初级生产者影响底栖食物网和碳流。该发现会改变珊瑚礁“生产者-消费者”结构的传统划分，并为气候变化下礁生态系统韧性提供新变量。

DOI: <https://doi.org/10.1038/s41559-026-03109-y>

环境与气候

17. Transport-constrained eddy-mean energetics sustain extreme rainfall over China

(输送约束的涡旋-平均能量过程维持中国极端降雨)

Communications Earth & Environment | 2026-05-22 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Extreme summer rainfall over China has intensified in recent decades, but the mechanisms organizing moisture transport and sustained ascent remain uncertain. This open-access study links transport-constrained eddy-mean energetics to the maintenance of extreme rainfall, offering physical insight for projections and adaptation planning in flood-prone regions.

中文摘要

研究关注中国夏季极端降雨增强背后的水汽输送和持续上升运动机制，提出输送约束下的涡旋-平均能量过程可维持极端降雨。该机制有助于解释长江、珠江等人口和经济密集区洪涝风险上升。

深度解读

极端降雨研究常停留在水汽增多和环流异常层面，而这项工作进一步追踪能量转换与输送约束。它对提高区域气候模式和灾害预警的物理一致性有意义。对生态环境管理而言，极端降雨会通过洪涝、滑坡、污染物再悬浮和河湖营养盐输入影响流域生态安全。

DOI: <https://doi.org/10.1038/s43247-026-03547-3>

环境与气候

18. A systematic overview and second-order meta-analysis of nature-based interventions for stress, anxiety and depression

(自然干预缓解压力、焦虑和抑郁的系统综述与二阶荟萃分析)

Nature Human Behaviour | 2026-05-28 | 【Recent】



图：Graphical abstract / article figure source: <https://www.nature.com/articles/s41562-026-02433-4/figures/1>

Abstract

A systematic overview and second-order meta-analysis covering 116 reviews and more than ten million estimated participants finds that nature-based interventions reduce negative mental-health outcomes and improve positive outcomes, while noting limitations in primary-study design and comparator choice.

中文摘要

研究整合 116 篇系统综述和超过一千万名估计参与者的证据，发现自然干预总体上可降低压力、焦虑和抑郁等负面结果，并提升积极情绪和放松水平。但研究也指出，原始研究中被动对照较多、干预定义不统一，限制了疗效比较。

深度解读

这篇文章把城市绿地、蓝绿空间和自然接触的公共健康价值推向更高证据层级。它对生态系统服务研究的意义在于，健康收益不再只是定性叙述，而可进入干预评估和政策成本收益分析。未来需要更严格的主动对照试验，区分自然本身、运动、社交和安静休息的相对贡献。

DOI: <https://doi.org/10.1038/s41562-026-02433-4>

环境与气候

19. Onset seasonality controls compound streamflow drought risk at a global scale

(起始季节性控制全球复合径流干旱风险)

npj Natural Hazards | 2026-05-21 | 【Recent】



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Streamflow droughts threaten water, food and energy security. This study defines compound streamflow droughts as low streamflow coinciding with or preceded by precipitation deficits and followed by elevated evaporative demand within two months of onset, and shows that onset seasonality is a key global control on compound drought risk.

中文摘要

研究将复合径流干旱定义为低径流与降水亏缺相伴或相继出现，并在起始前后两个月内叠加较高蒸发需求的事件。结果表明，干旱起始季节性是控制全球复合径流干旱风险的重要因素。

深度解读

传统干旱指标往往把降水、径流和蒸发需求分开处理，而复合干旱更接近水资源系统承受的真实压力。起始季节决定植被需水、蓄水状态和人类用水峰值，因此会放大或缓冲生态影响。流域管理需要把季节性预警转化为水库调度、生态流量和农业用水的动态规则。

植物与农业

20. A transcriptional atlas of early Arabidopsis seed development suggests mechanisms for inter-tissue coordination

(拟南芥早期种子发育转录图谱揭示组织间协调机制)

Nature Plants | 2026-05-21 | **【Recent】**



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Successful seed development in flowering plants requires coordination among the embryo, endosperm and maternal seed coat. This resource article presents a transcriptional atlas of early Arabidopsis seed development and suggests mechanisms for inter-tissue coordination during reproduction.

中文摘要

被子植物种子发育需要胚、胚乳和母体种皮三个遗传来源不同的组织协调。该资源型研究构建拟南芥早期种子发育的转录图谱，并提出组织间协调发育的潜在机制。

深度解读

虽然该文偏植物发育基础研究，但对农业生态系统中的种子产量、逆境响应和繁殖成功具有基础意义。高分辨率转录图谱可帮助识别胚乳、种皮和胚之间的信号通路。长期看，这类资源会支持作物育种中对种子大小、灌浆和环境适应性的精准调控。

DOI: <https://doi.org/10.1038/s41477-026-02295-8>

环境技术与可持续

21. Upgrading glycerol to sorbose via a tandem photoelectrocatalysis-enzyme catalysis relay

(串联光电催化-酶催化将甘油升级为山梨糖)

Nature Sustainability | 2026-05-28 | **【Recent】**



图：Graphical abstract / article figure source: <https://www.nature.com/articles/s41893-026-01856-1/figures/1>

Abstract

The study develops a tandem photoelectrocatalysis-enzyme catalysis route to convert low-cost glycerol, a biodiesel by-product, into sorbose, a higher-value sugar used in food and pharmaceutical applications. It illustrates a circular biorefinery pathway for upgrading waste carbon streams.

中文摘要

研究提出光电催化与酶催化串联策略，将生物柴油副产低值甘油转化为高值山梨糖。该方法为废弃或低值碳流的精细化利用提供了可持续化学路径。

深度解读

该文偏环境技术，但与循环经济和生物质资源化高度相关。将副产甘油升级为高值化合物，有助于提升生物柴油产业链的整体环境绩效。它也体现了可持续制造从“处理废物”走向“重构碳流价值”的趋势。

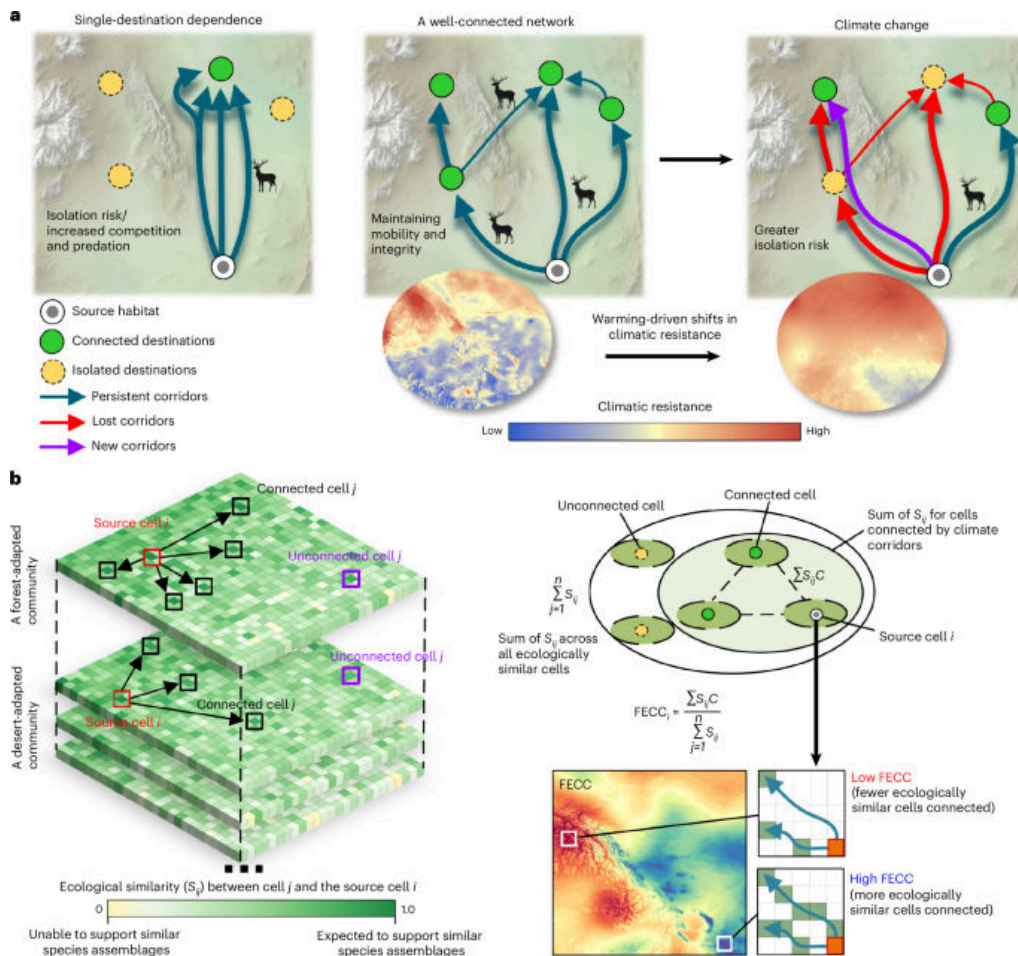
DOI: <https://doi.org/10.1038/s41893-026-01856-1>

环境与气候

22. Warming erodes climate connectivity for terrestrial vertebrates

(变暖削弱陆生脊椎动物的气候连通性)

Nature Climate Change | 2026-05-18 | 【Recent】



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Climate Change, Published online: 18 May 2026; doi:10.1038/s41558-026-02658-1 The authors consider connectivity—specifically, the functional effectiveness of climate connectivity (FECC)—under climate change. Under high emissions (SSP5-8.5), FECC is projected to decline across 77% of the global land area by 2061–2080, risking ecological isolation and subsequent biodiversity loss.

中文摘要

作者从气候连通性的功能有效性（FECC）出发，评估气候变化下陆生脊椎动物沿适宜气候通道迁移的可能性。在高排放情景 SSP5-8.5 下，到 2061—2080 年，全球约 77% 陆地区域的 FECC 预计下降，增加生态隔离和生物多样性损失风险。

深度解读

这项研究把保护区和栖息地连通性从静态空间格局推进到动态气候适宜性层面。其核心意义在于指出，仅保留现有栖息地斑块并不足以保障物种随气候迁移。对保护规划而言，廊道设计需要优先考虑未来气候通道的持续性，而非只依据当前物种分布。

DOI: <https://doi.org/10.1038/s41558-026-02658-1>

环境与气候

23. Seabird range contraction and dispersal under climate change

(气候变化下海鸟分布范围收缩与扩散距离增加)

Nature Climate Change | 2026-05-19 | 【Recent】

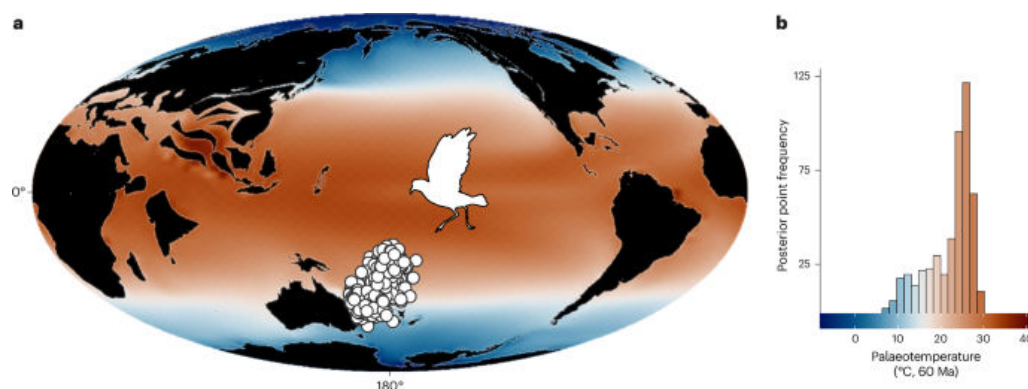


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Climate Change, Published online: 19 May 2026; doi:10.1038/s41558-026-02655-4 The authors reconstruct historical seabird dispersal routes, showing that birds responded to temperature shifts by changes in range size rather than body mass. These trends are projected to persist, with higher rates of warming causing greater range contractions and longer dispersal distances.

中文摘要

作者重建了历史海鸟扩散路线，发现海鸟对温度变化的响应主要表现为分布范围大小变化，而不是体重变化。模型预计这一趋势将延续，升温速率越高，分布范围收缩越明显，扩散距离也越长。

深度解读

海鸟具有较强移动能力，但这并不意味着它们能轻松逃避气候压力。研究显示，温度变化会重塑适宜栖息地的空间结构，使种群面临更远距离扩散和繁殖地丧失的双重压力。对海洋保护区而言，固定边界可能难以覆盖未来关键栖息地。

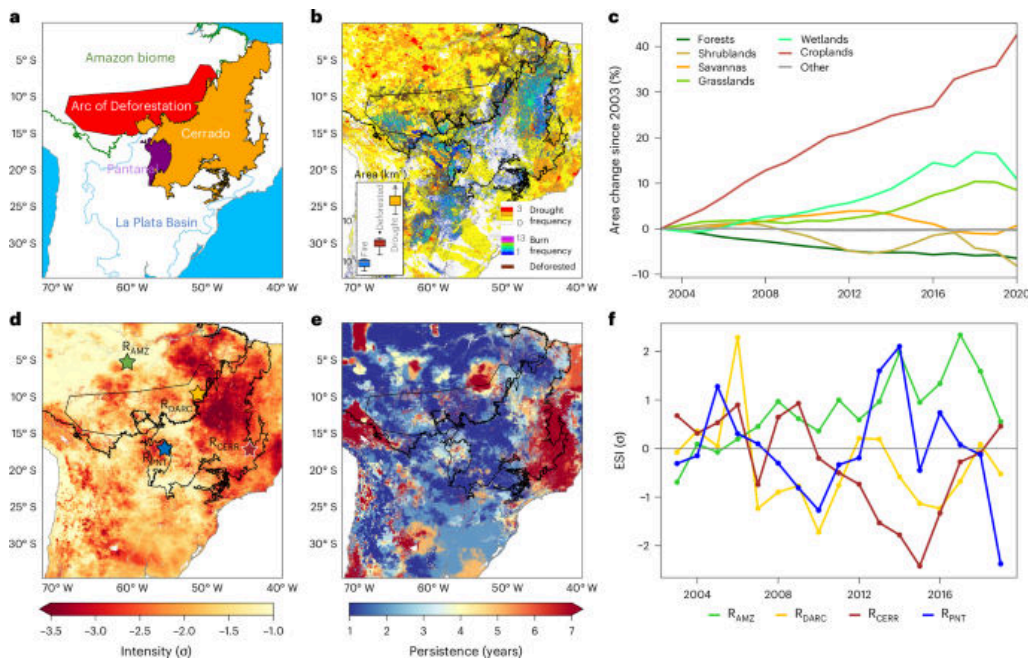
DOI: <https://doi.org/10.1038/s41558-026-02655-4>

环境与气候

24. Evapotranspiration declines prolonged by deforestation and fire in South American biomes

(南美生物群系中森林砍伐与火灾延长蒸散下降过程)

Nature Geoscience | 2026-05-19 | 【Recent】



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Geoscience, Published online: 19 May 2026; doi:10.1038/s41561-026-01981-8 Human-driven disturbances intensify and prolong evapotranspiration declines across South American ecosystems, disrupting the water balance and threatening ecological resilience, according to hydrological modelling and remote sensing data.

中文摘要

基于水文模型和遥感数据，研究表明人类驱动的扰动会加剧并延长南美生态系统的蒸散下降，扰乱区域水量平衡并威胁生态韧性。森林砍伐和火灾是影响植被—水循环耦合的重要压力源。

深度解读

蒸散是陆地生态系统调节区域气候和水循环的关键通量。扰动后蒸散长期偏低意味着植被恢复、降水反馈和热环境调节都可能受到持续影响。该结果支持将控火、减少毁林和生态恢复作为水安全政策的一部分。

DOI: <https://doi.org/10.1038/s41561-026-01981-8>

环境与气候

25. Personal experiences matter for climate action

(个人经历对气候行动至关重要)

Nature Climate Change | 2026-05-19 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Climate Change, Published online: 19 May 2026; doi:10.1038/s41558-026-02640-x Climate change impacts are no longer distant but have entered people's everyday experiences. Here we look back on a 2011 paper that showed how direct personal experience shapes people's climate change perceptions, their beliefs about the efficacy of their action and willingness to act, and how the field of research has evolved.

中文摘要

文章回顾一篇 2011 年研究，讨论直接个人经历如何塑造公众对气候变化的感知、对自身行动有效性的信念以及行动意愿。随着气候影响进入日常生活，该领域研究也在持续发展。

深度解读

气候传播不再只是传递抽象风险，而要理解人们如何从亲身经历中形成判断。个人经历能提高风险感知，但是否转化为行动还取决于效能感、社会规范和可行选择。该文对气候沟通和社区适应动员具有现实意义。

DOI: <https://doi.org/10.1038/s41558-026-02640-x>

土壤与生物地球化学

26. The efficiency and ocean acidification mitigation potential of ocean alkalinity enhancement on multi-centennial timescales

(多世纪尺度上海洋碱度增强的效率及缓解海洋酸化潜力)

Biogeosciences | 2026-05-18 | **【Recent】**



图：Graphical abstract / article figure source:

Abstract

The efficiency and ocean acidification mitigation potential of ocean alkalinity enhancement on multi-centennial timescales Hendrik Grosselindemann, Friedrich A. Burger, and Thomas L. Frölicher *Biogeosciences*, 23, 3299–3321, <https://doi.org/10.5194/bg-23-3299-2026>, 2026 We assess the long-term carbon cycle, climate and ocean acidification response of ocean alkalinity enhancement (OAE) using an emission-driven Earth system model across warming stabilization scenarios. OAE lowers atmospheric CO₂ and produces a scenario-independent linear cooling. Gross carbon capture efficiency remains high, while net efficiencies decline over time due to carbon-cycle feedbacks. Ocean acidification mitigation by OAE is dominated by CO₂ drawdown on long timescales.

中文摘要

研究使用排放驱动的地球系统模型，在不同变暖稳定情景下评估海洋碱度增强（OAE）对长期碳循环、气候和海洋酸化的影响。结果显示，OAE 可降低大气 CO₂ 并产生相对情景独立的线性降温，总碳捕获效率较高，但净效率会因碳循环反馈随时间下降；长期尺度上海洋酸化缓解主要由 CO₂ 抽降主导。

深度解读

该研究把 OAE 从短期碳移除概念推进到多世纪反馈评估。总效率与净效率的差异说明，海洋碳汇工程必须考虑再平衡和反馈过程，而不能只看初始吸收。其结论有助于约束 OAE 的气候收益、酸化缓解收益和长期治理风险。

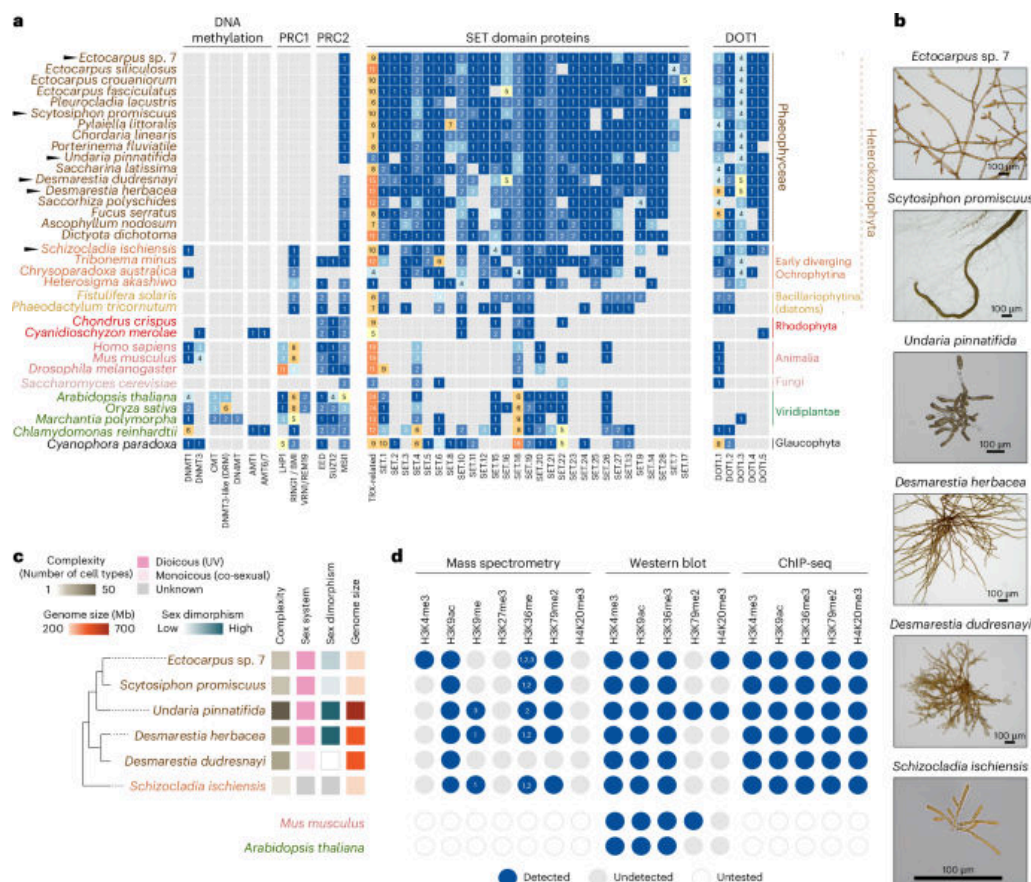
DOI: <https://doi.org/10.5194/bg-23-3299-2026>

土壤与生物地球化学

27. Marine carbonate system variability from tidal to seasonal timescales at the interface between the North Sea and Wadden Sea

(北海与瓦登海交界处从潮汐到季节尺度的海洋碳酸盐体系变异)

Biogeosciences | 2026-05-18 | **【Recent】**



图：Graphical abstract / article figure source:

Abstract

Marine carbonate system variability from tidal to seasonal timescales at the interface between the North Sea and Wadden Sea Yasmina Ourradi, Gert-Jan Reichart, Sonja van Leeuwen, and Matthew P. Humphreys Bio-geosciences, 23, 3323–3346, <https://doi.org/10.5194/bg-23-3323-2026>, 2026 Coastal seas play a key role in the global carbon cycle, yet what drives their carbonate chemistry remains poorly understood. We measured pH at high frequency for a year at the Wadden Sea-North Sea interface. Biological activity drives pH changes, while freshwater inflow shapes total alkalinity. Both processes influence dissolved inorganic carbon. The Wadden Sea acts as a net carbon dioxide source to the atmosphere. High-frequency monitoring is essential under a changing climate.

中文摘要

研究在瓦登海—北海交界处进行一年高频 pH 观测，分析海岸海域碳酸盐化学的驱动因素。结果表明，生物活动主导 pH 变化，淡水输入影响总碱度，两者共同影响溶解无机碳；瓦登海表现为大气 CO₂ 的净源。

深度解读

海岸海域碳循环具有强烈的高频波动，低频采样容易遗漏关键过程。生物生产、呼吸和淡水输入共同控制碳酸盐体系，使酸化风险呈现潮汐到季节尺度的变化。该研究凸显在气候变化下建立高频监测网络的必要性。

DOI: <https://doi.org/10.5194/bg-23-3323-2026>

28. Towards sustainable metal–organic frameworks from lab to practice
(推动金属—有机框架材料从实验室走向实践的可持续性评估)

Nature Sustainability | 2026-05-19 | 【Recent】

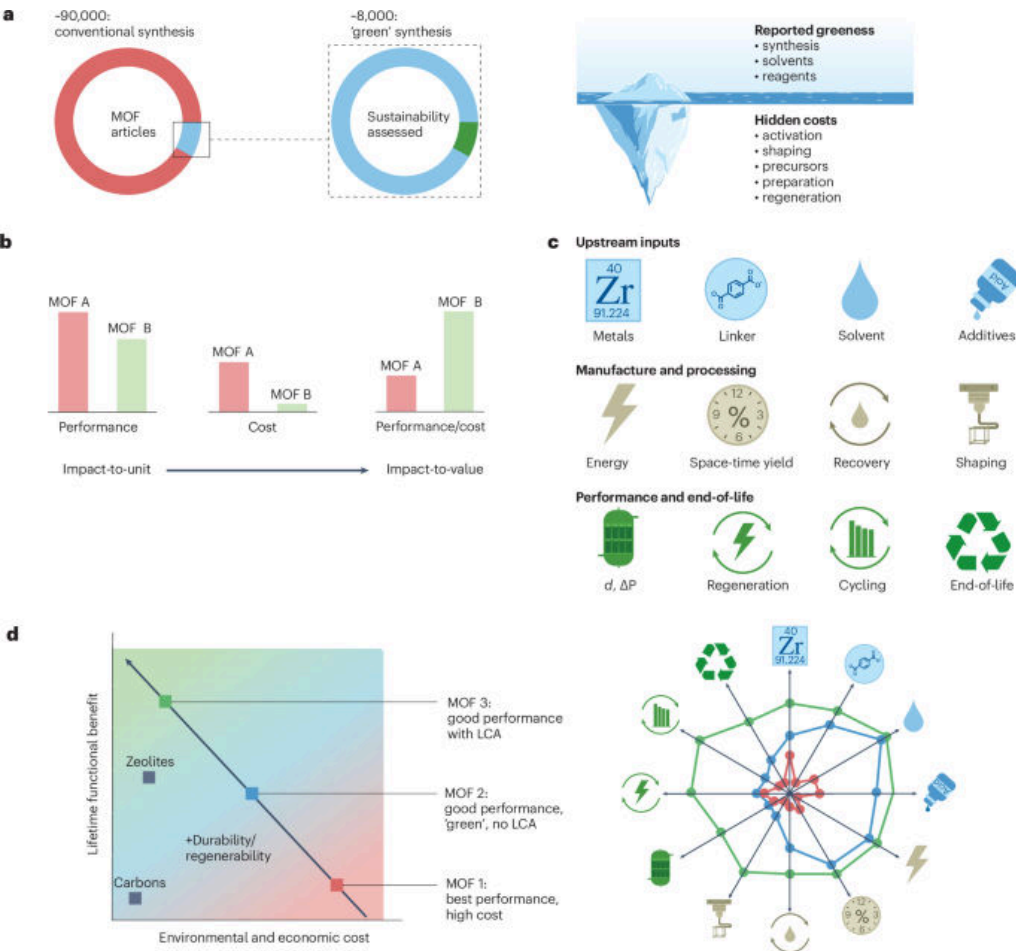


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Sustainability, Published online: 19 May 2026; doi:10.1038/s41893-026-01839-2Realistic life-cycle impacts of metal–organic frameworks (MOFs) are rarely quantified, despite their promise for carbon capture, water treatment, catalysis and other environmental applications. Here we propose a compact, actionable sustainability toolkit for MOFs to systematically verify claims of ‘green’.

中文摘要

金属—有机框架材料（MOFs）在碳捕集、水处理和催化等环境应用中具有潜力，但其真实生命周期影响很少被量化。作者提出一套紧凑、可操作的 MOF 可持续性工具包，用于系统核验“绿色”声明。

深度解读

新材料的环境价值不能只由单项性能指标决定，还必须纳入原料、合成、能耗、寿命和末端处置。MOF 若要规模化应用，需要证明其全生命周期收益大于替代技术。该文为材料研发早期嵌入可持续性筛选提供了方法框架。

DOI: <https://doi.org/10.1038/s41893-026-01839-2>

29. Emergent constraints on future methane emissions from global wetlands
(全球湿地未来甲烷排放的涌现约束)

Nature Geoscience | 2026-05-19 | 【Recent】

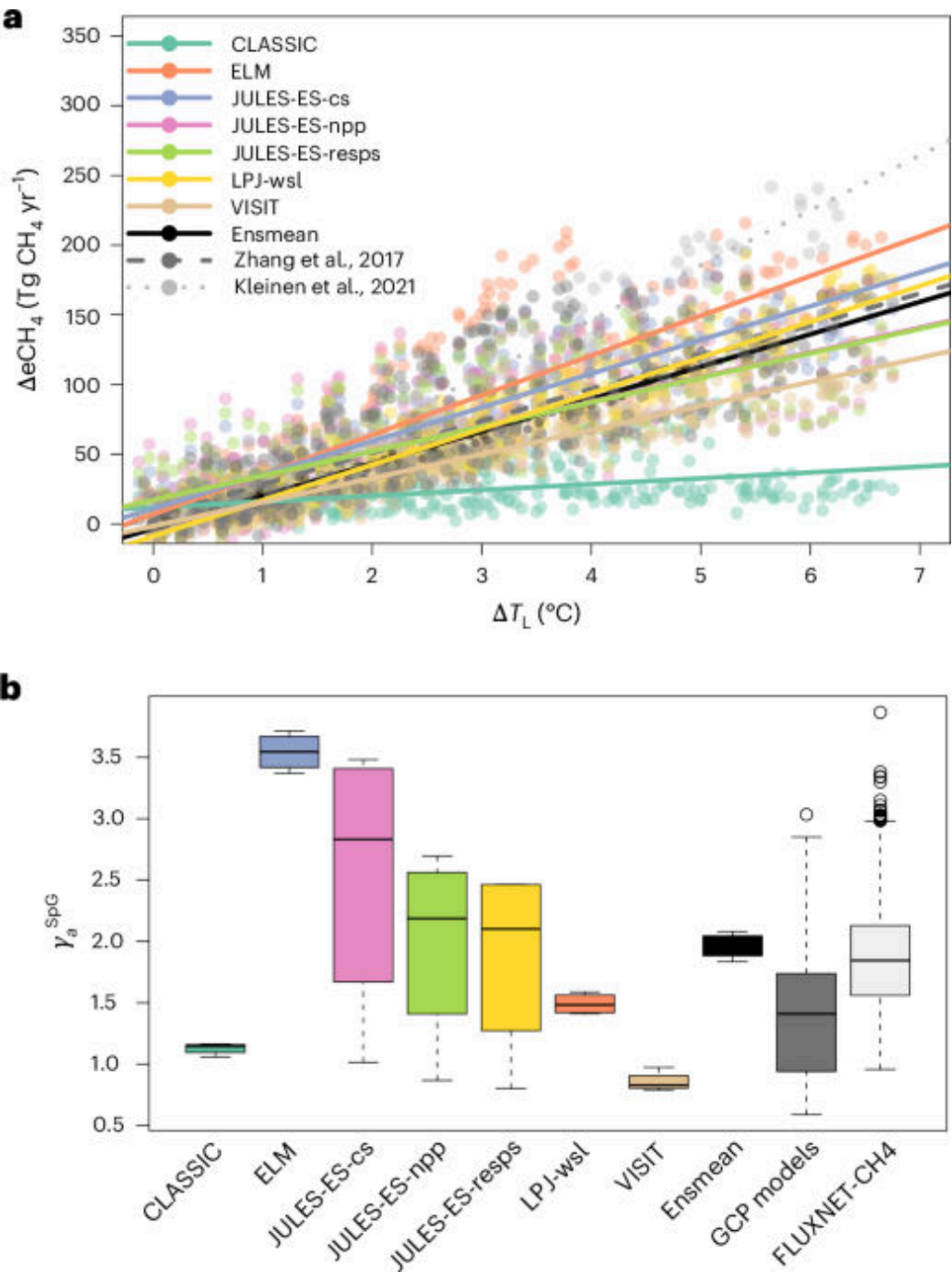


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Geoscience, Published online: 19 May 2026; doi:10.1038/s41561-026-01987-2Enhanced future methane emissions from global wetlands under warming could substantially offset the emissions reduction goals of the Global Methane Pledge, according to ensemble simulations from terrestrial biosphere models.

中文摘要

基于陆地生物圈模型集合模拟，研究表明变暖下全球湿地未来甲烷排放增强，可能显著抵消《全球甲烷承诺》的减排目标。该文利用涌现约束思路缩小未来排放不确定性。

深度解读

湿地甲烷是气候—碳循环反馈中的关键不确定源。变暖会通过温度、湿地面积、水位和底物供应改变甲烷产生与释放。若自然源增加被低估，人为甲烷减排目标需要更强力度才能实现同等气候效果。

DOI: <https://doi.org/10.1038/s41561-026-01987-2>

环境与气候

30. Synergies between poverty reduction and biodiversity conservation

(减贫与生物多样性保护之间的协同关系)

Nature Sustainability | 2026-05-18 | 【Recent】

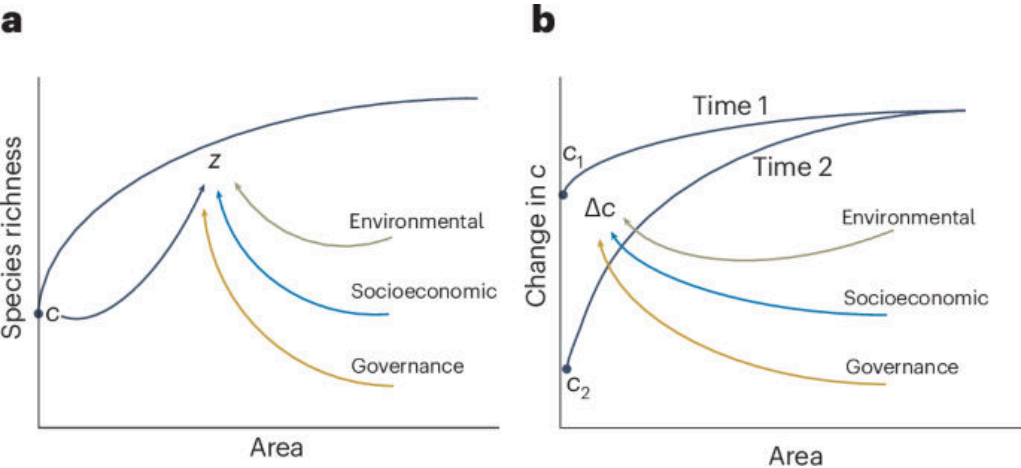


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Sustainability, Published online: 18 May 2026; doi:10.1038/s41893-026-01830-x Greater household dependence on forest resources is associated with lower tree species diversity in tropical forests, highlighting the interconnected dynamics of poverty and biodiversity in multifunctional landscapes. Forest policies that aim to conserve biodiversity must consider the interconnected dynamics of poverty alleviation and biodiversity conservation to achieve multiple sustainability goals.

中文摘要

研究发现，家庭对森林资源依赖程度越高，热带森林树种多样性往往越低，凸显多功能景观中贫困与生物多样性的相互联系。森林政策若要保护生物多样性，需要同时考虑减贫和保护目标。

深度解读

该研究避免把当地居民生计与保护目标简单对立，而是强调二者之间的结构性联系。高资源依赖可能反映替代生计不足、制度支持不足或市场压力。有效保护需要通过减贫、权益安排和可持续利用降低对森林的高强度依赖。

DOI: <https://doi.org/10.1038/s41893-026-01830-x>

环境与气候

31. Advancing biodiversity through poverty solutions (通过贫困解决方案促进生物多样性)

Nature Sustainability | 2026-05-18 | 【Recent】

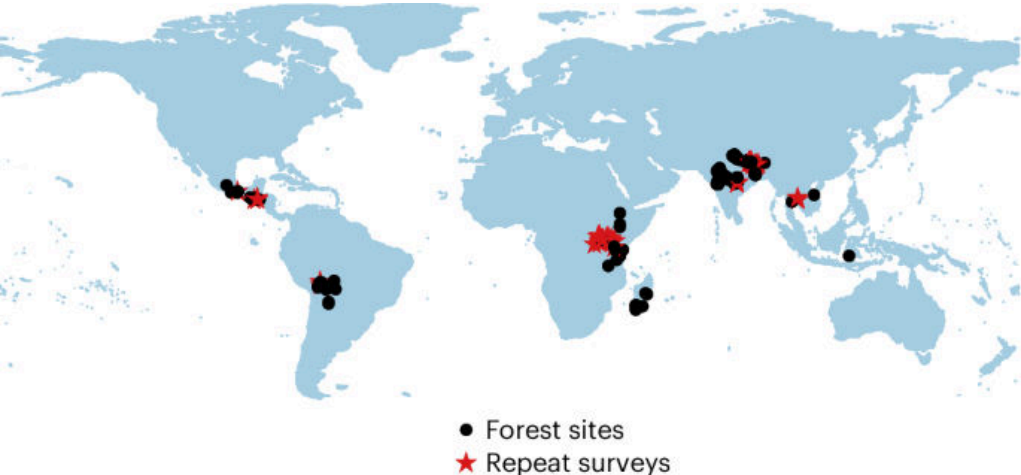


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Sustainability, Published online: 18 May 2026; doi:10.1038/s41893-026-01816-9 Forest biodiversity is linked, in many cases, to the myriad ecosystem services that support the livelihoods of human populations. Authors here examine the demographic, socioeconomic and institutional variables that impact forest biodiversity across 322 tropical forests.

中文摘要

文章指出，森林生物多样性在许多情况下与支持人类生计的多种生态系统服务相联系。作者考察了322个热带森林中人口、社会经济和制度变量对森林生物多样性的影响。

深度解读

这项研究把热带森林保护置于社会—生态系统框架下，而非单纯生态变量解释。人口结构、收入来源和制度安排会改变森林利用强度及其对多样性的影响。它提示保护政策需要与社会治理和生计改善同步设计。

DOI: <https://doi.org/10.1038/s41893-026-01816-9>

环境与气候

32. Sea-level-driven land conversion amplified by coastal agriculture (沿海农业放大海平面驱动的土地转换)

Nature Sustainability | 2026-05-18 | 【Recent】

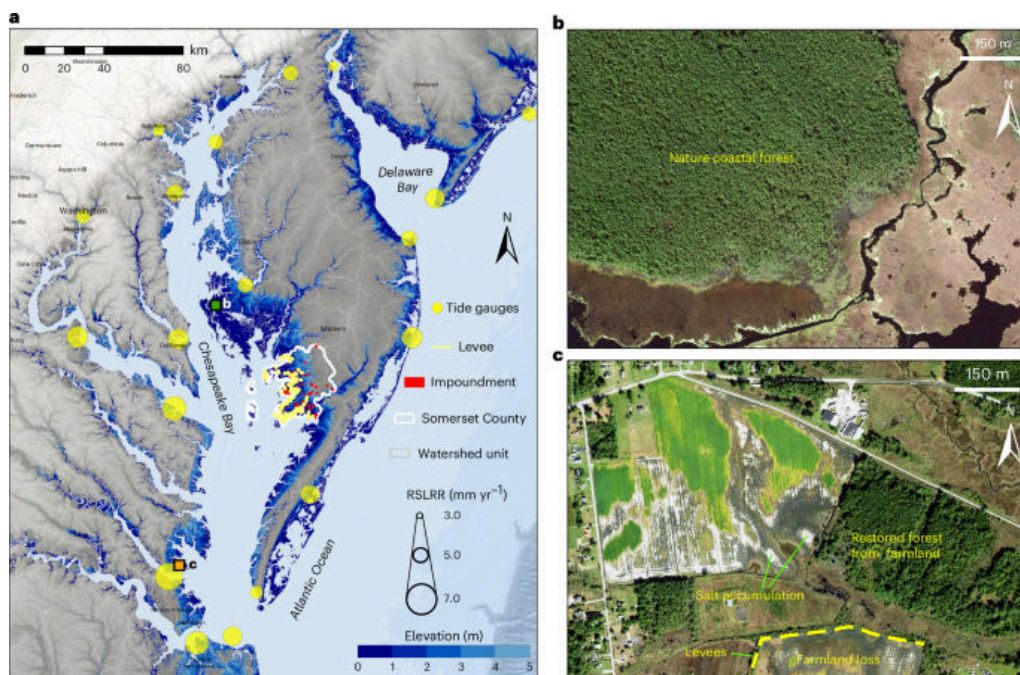


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Sustainability, Published online: 18 May 2026; doi:10.1038/s41893-026-01835-6As sea levels rise, coastal ecosystems and economies are threatened by saltwater intrusion and wetland encroachment. These effects are accelerated by coastal agriculture, highlighting the importance of local management interventions.

中文摘要

随着海平面上升，盐水入侵和湿地扩张威胁沿海生态系统与经济活动。研究指出，沿海农业会加速这些影响，凸显地方管理干预的重要性。

深度解读

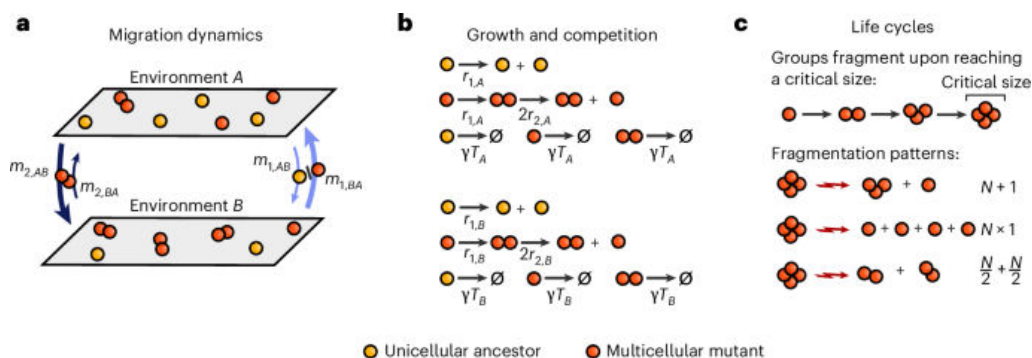
海平面上升的土地影响会被本地土地利用方式显著调节。农业排水、地表改造和淡水管理可能降低系统抵抗盐水入侵与湿地迁移的能力。沿海适应需要把农田管理、湿地缓冲和有序退让结合起来。

DOI: <https://doi.org/10.1038/s41893-026-01835-6>

环境与气候

33. [ASAP] Remote Sensing Enables Basin-Scale Inventories of Coal Mine Methane
(遥感支持流域尺度煤矿甲烷清单构建)

Environmental Science & Technology | 2026-05-19 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Environmental Science & Technology DOI: 10.1021/acs.est.5c14976

中文摘要

该 Environmental Science & Technology ASAP 论文题名显示，研究关注利用遥感技术开展流域尺度煤矿甲烷排放清单。所给摘要仅提供期刊和 DOI 信息，具体遥感平台、反演方法和排放结果需以原文为准。

深度解读

煤矿甲烷是能源系统中重要且可减排的温室气体来源。遥感能够弥补自报清单在空间覆盖和高排放点识别上的不足。流域尺度清单有助于把排放监测从设施级发现推进到区域治理和政策核查。

DOI: <https://doi.org/10.1021/acs.est.5c14976>

植物与农业

34. Agri-environmental policies have reduced cropland degradation globally

(农业环境政策已在全球范围降低耕地退化)

Nature Food | 2026-05-18 | 【Recent】

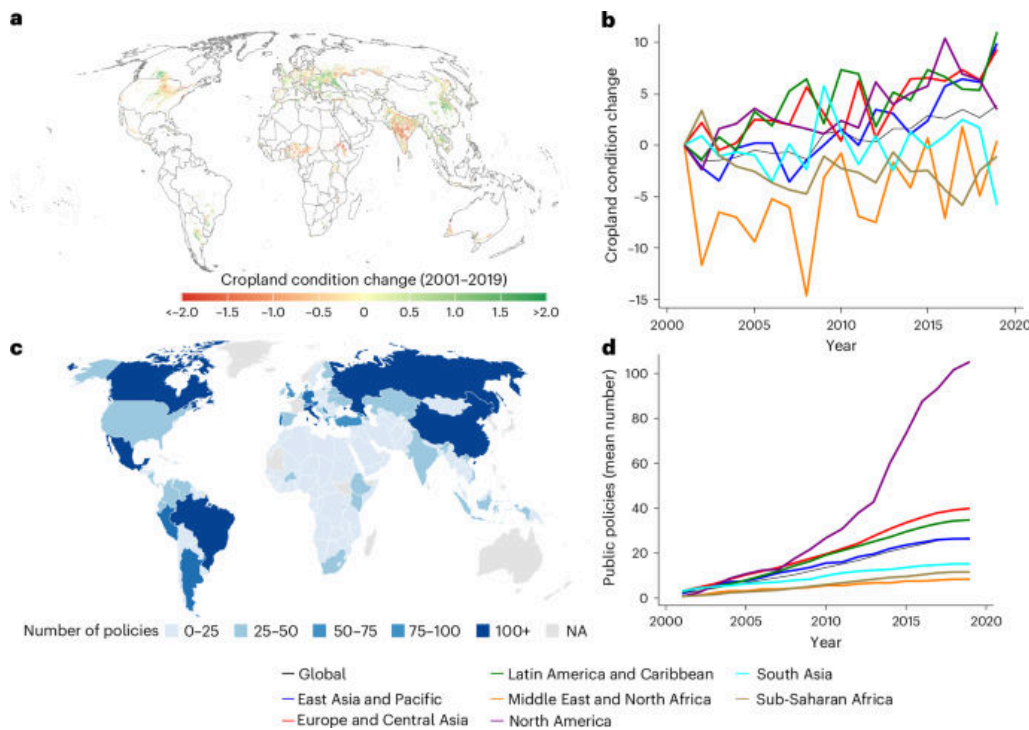


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Food, Published online: 18 May 2026; doi:10.1038/s43016-026-01359-4 Evidence on the effectiveness of national policies against cropland degradation is limited. Using global satellite data from 2001 to 2019, this study addresses this gap and reveals which country characteristics are associated with distinct policy success levels.

中文摘要

关于国家政策抑制耕地退化有效性的证据仍有限。该研究利用 2001—2019 年全球卫星数据评估政策效果，并识别与不同政策成功水平相关的国家特征。

深度解读

该研究为农业环境政策提供了全球尺度的遥感证据。耕地退化受气候、管理、市场和治理能力共同影响，政策效果也会因国家条件而异。结果可帮助从“是否有政策”转向评估“什么条件下政策有效”。

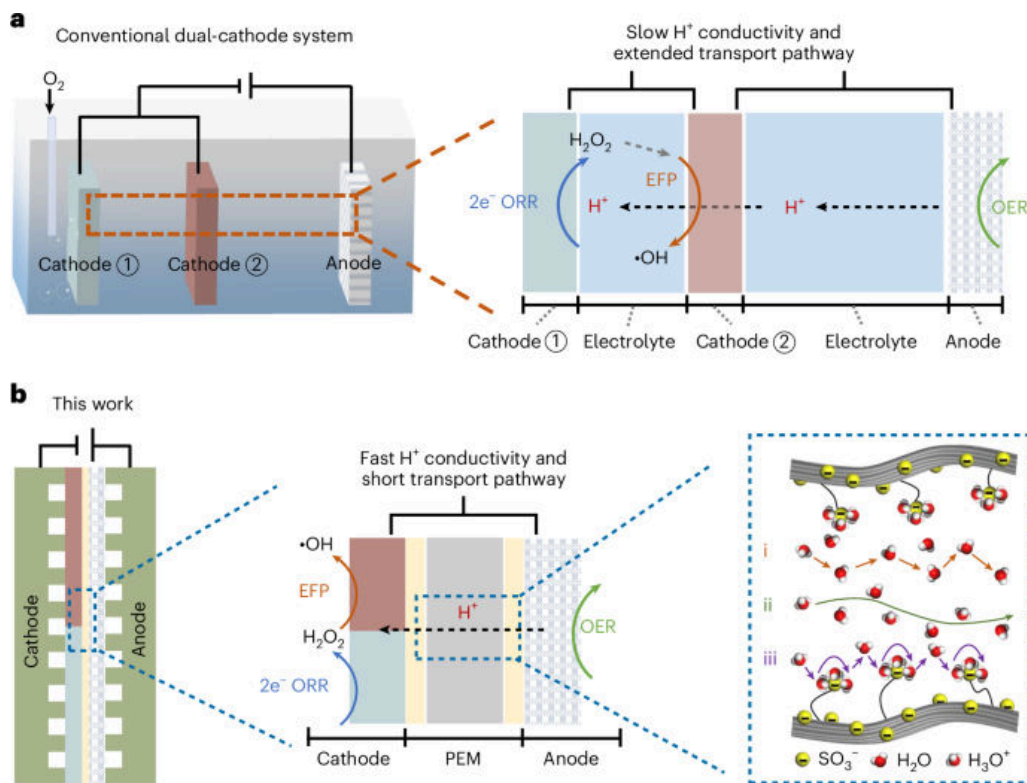
DOI: <https://doi.org/10.1038/s43016-026-01359-4>

遥感与模型

35. Reducing Spatial Bias at Heterogeneous Sites: A Novel Eddy Covariance Filtering Approach Integrating Footprint Modeling and Remote Sensing

(降低异质性站点空间偏差：融合通量足迹模型与遥感的新型涡度协方差筛选方法)

Journal of Geophysical Research - Biogeosciences | 2026-05-18 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 5, May 2026.

中文摘要

该 Journal of Geophysical Research: Biogeosciences 论文题名表明，研究提出一种结合通量足迹建模和遥感的新型涡度协方差数据筛选方法，以降低异质性站点的空间代表性偏差。所给摘要仅含期刊卷期信息，具体算法和验证结果需以原文为准。

深度解读

涡度协方差观测常被用于估算生态系统碳水通量，但在土地覆盖异质区域容易受足迹变化影响。将遥感空间信息与足迹模型结合，可更清楚地识别通量观测实际代表的地表单元。该方法有助于提高站点观测与区域尺度模型或卫星产品比较的可靠性。

DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009619>